

Recyclable Item Classification Using Convolutional Neural Networks and Transfer Learning

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Abstract

This paper presents a comparative study of two deep learning approaches for automatically classifying recyclable waste into four material categories: glass, metal, paper, and plastic. The growing challenge of effective waste sorting motivates the need for intelligent, automated solutions that can assist both industrial and consumer-level recycling efforts. We trained and evaluated a custom Convolutional Neural Network (CNN) and a transfer-learning model using the pretrained ResNet50 architecture on a dataset of 12,873 images resized to 150x150 pixels. Both models were trained for 10 epochs using the Adam optimizer. The ResNet50 transfer learning model achieved a test accuracy of 87.90% substantially outperforming the custom CNN at 61.83%. These results demonstrate that leveraging pre-trained features from large-scale datasets significantly improves classification performance even without fine-tuning base model.

Introduction

Improper disposal of recyclable materials is a major contributor to environmental pollution and landfill overflow. Manual sorting of waste is slow, costly, and error-prone, creating strong motivation for automated classification systems capable of identifying recyclable items quickly and accurately.

Recent advances in deep learning and convolutional neural networks (CNNs) have transformed computer vision, enabling machines to classify images with near-human accuracy on many benchmarks. These capabilities offer a promising path forward for waste classification: given an image of an item, a trained model can predict its material category and guide proper disposal.

This project investigates whether a custom CNN trained from scratch can perform competitively compared to a transfer-learning approach using ResNet50, a deep architecture pre-trained on the large-scale ImageNet dataset. We use a dataset of four recyclable material classes and evaluate both models under identical conditions to enable a fair comparison. The remainder of this paper reviews related work (Section 2), describes the dataset and models (Section 3), reports experimental results (Section 4), and concludes with a summary and future directions (Section 5).

Related Work

Automated waste classification has attracted significant research attention from the computer vision community in recent years. One of the most widely cited early contributions is the TrashNet dataset introduced by Thung and Yang (2016), which contains approximately 2,527 images across six categories and served as a benchmark for initial CNN-based waste classifiers. Their work demonstrated that even relatively shallow CNNs could achieve reasonable accuracy on this task.

Subsequent studies explored more powerful architectures. Awe et al (2017) demonstrated that fine-tuning VGG-16, pre-trained on ImageNet, yielded significantly higher accuracy than training from scratch, suggesting the advantage of transfer learning when labeled data is limited. Mittal et al. (2019) compared multiple pre-trained architectures including InceptionV3 and MobileNet, finding that transfer learning consistently outperformed custom CNNs on moderate sized datasets.

ResNet50, the architecture used in this work, was introduced by He et al. (2016) and achieves strong performance through residual connections that facilitate training of very deep networks. It has been widely applied as a feature extractor for downstream vision tasks, making it a natural candidate for recyclable waste classification. Our work provides a controlled comparison of a custom CNN against ResNet50 transfer learning under identical training conditions.

Methodologies

Dataset

We used a dataset of recyclable item images organized into four categories: glass, metal, paper, and plastic. The full dataset contains 12,873 images split into training (9,009 images, 70%), validation (2,575 images, 20%), and test (1,289 images, 10%) sets using a random shuffled split per class. All images were resized to 150x150 pixels with three color channels (RGB) and loaded in batches of 32 using TensorFlow's `image_dataset_from_directory` utility.

Preprocessing and Data Pipeline

For the custom CNN, pixel values were normalized to the $[0, 1]$ range using a Rescaling ($1./255$) layer as the first stage of the model. For the ResNet50 model, the ImageNet-specific preprocess input function was applied as a preprocessing layer, which rescales and mean-subtracts inputs according to the statistics used during ResNet50 pretraining. Both datasets were cached and prefetched using TensorFlow's AUTOTUNE setting to maximize training throughput.

Model 1: Custom CNN Architecture

The custom CNN is a Sequential model consisting of three convolutional blocks, each containing a Conv2D layer followed by a MaxPooling2D layer. The convolutional layers use ReLU activations and progressively increase the number of filters: 32, 64, and 128, with 3×3 kernels throughout. After the three convolutional blocks, the feature maps are flattened and passed into a Dense layer of 128 units with ReLU activation, followed by a Dropout layer with rate 0.5 to reduce overfitting. The output is a Dense layer with 4 units and softmax activation. The model contains 4,828,868 trainable parameters and was compiled with the Adam optimizer and sparse categorical cross-entropy loss.

Model 2: ResNet50 Transfer Learning

The second model uses ResNet50 pre-trained on ImageNet as a frozen feature extractor. The base model was loaded with `include_top=False` and its trainable attribute set to `False`, so no base weights were updated during training. On top of the frozen base, a GlobalAveragePooling2D layer reduces the spatial feature maps to a single vector, followed by a Dense layer of 128 units with ReLU activation, a Dropout layer with rate 0.5, and a final Dense output layer with 4 units and softmax activation. The model was compiled and trained using the same optimizer, loss function, and number of epochs as the custom CNN.

Experiments and Results

Training Behavior

Both models were trained for 10 epochs. The custom CNN showed a steady increase in training accuracy, reaching 81.50% by epoch 10, while validation accuracy plateaued at 62.83%. This gap is a clear sign of overfitting: the model memorized training examples but failed to generalize to unseen data.

The ResNet50 transfer learning model exhibited more consistent and stable training. Starting from 74.03% training accuracy in epoch 1, the model converged to 92.51%

training accuracy and 87.88% validation accuracy by epoch 10. The small gap between training and validation accuracy suggests that the pre-trained features generalized well to the recyclable waste domain, with the frozen base effectively acting as a strong regularizer.

Test Set Evaluation

The custom CNN finished training with a final test accuracy of 61.83% and a validation accuracy of 62.83%, while the ResNet50 transfer learning model achieved a test accuracy of 87.90% and a validation accuracy of 87.88%. This represents a difference of over 26 percentage points on the test set, clearly favoring the transfer learning approach. The close agreement between validation and test accuracy for both models confirms that the validation split served as a reliable performance estimate.

Analysis

The large performance gap between the two models can be attributed to the depth and representational capacity of ResNet50's pre-trained weights. ResNet50 was trained on over one million images across 1,000 categories and has learned rich, general-purpose visual features — such as edges, textures, and object parts — that transfer directly to material classification tasks. The custom CNN, by contrast, must learn all its representations from roughly 9,000 training images, which is insufficient to develop the hierarchical feature detectors needed for robust classification.

The overfitting observed in the custom CNN is consistent with this explanation. The dropout layer mitigated but did not fully prevent the model from over-relying on training-specific patterns. Future work could address this by introducing data augmentation or L2 weight regularization.

Conclusions

This project investigated two deep learning models for classifying images of recyclable waste into four material categories: glass, metal, paper, and plastic. A custom CNN trained from scratch and a ResNet50 model with frozen pre-trained weights were both trained for 10 epochs under identical conditions on a dataset of 12,873 images resized to 150×150 pixels.

The results clearly favor the transfer learning approach. The ResNet50 model achieved 87.90% test accuracy with stable training and minimal overfitting, while the custom CNN reached only 61.83% and showed significant signs of overfitting. These findings are consistent with the broader literature demonstrating that transfer learning from large pre-trained models provides a substantial advantage when task-specific training data is limited.

Future directions include applying data augmentation to improve the custom CNN's generalization; fine-tuning the upper layers of ResNet50 with a low learning rate; exploring lightweight architectures such as MobileNetV2 for deployment on embedded devices; and expanding the dataset to include additional categories such as cardboard or e-waste.

References

- Awe, O.; Mengistu, R.; and Sreedhar, V. 2017. Smart Trash Net: Waste Localization and Classification. CS229 Project Report. Stanford, CA: Stanford University.
- He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 770–778.
- Mittal, G.; Yagnik, K. B.; Garg, M.; and Krishnamurthy, C. V. 2016. SpotGarbage: Smartphone App to Detect Garbage Using Deep Learning. In Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing, 940–945.
- Simonyan, K., and Zisserman, A. 2014. Very Deep Convolutional Networks for Large-Scale Image Recognition. arXiv preprint. arXiv:1409.1556. Ithaca, NY: Cornell University Library.
- TensorFlow Developers. 2024. TensorFlow: A System for Large-Scale Machine Learning. tensorflow.org. Accessed: 2024-01-01.
- Thung, G., and Yang, M. 2016. Classification of Trash for Recyclability Status. CS229 Project Report. Stanford, CA: Stanford University.
- Yang, M., and Thung, G. 2016. Classification of Trash for Recyclability Status Using Deep Learning Features. CS229 Project Report. Stanford, CA: Stanford University.